

	DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE Supplementary Examination – Summer 2023 Course: B. Tech. (Common to all Branches) Semester : III Subject Name & Code: Engineering Mathematics – III (BTBSC 301) Max Marks: 60 Date:08/08/2023 Duration: 3 Hrs.	
	Instructions to the Students: 1. Attempt any FIVE of the following questions. 2. All questions carry equal marks. 3. Use of non-programmable scientific calculators is allowed. 4. Assume suitable data wherever necessary and mention it clearly.	
		Marks
Q. 1	Solve Any Two of the following.	12
A)	Find the Laplace transform of $F(t) = \frac{e^{-at} - e^{-bt}}{t}$	6
B)	Find the Laplace transform of $F(t) = \sin 2t \cos 3t$	6
C)	Find the Laplace transform of $\operatorname{erf}(\sqrt{t})$.	6
Q.2	Solve Any Two of the following:	12
A)	State and prove the convolution theorem for finding the inverse Laplace transform.	6
B)	Using Partial Fraction method, find the inverse Laplace transform of $\bar{f}(s) = \frac{5s+3}{(s-1)(s^2+2s+5)}$	6
C)	Find the inverse Laplace transform of $\bar{f}(s) = \cot^{-1}\left(\frac{s+3}{2}\right)$	6
Q. 3	Solve any Two of the following:	12
A)	Find the Fourier sine transform of $e^{- x }$, and hence show that $\int_0^\infty \frac{x \sin mx}{1+x^2} dx = \frac{\pi e^{-m}}{2}$, $m > 0$	6
B)	Find the Fourier transform of $f(x) = \begin{cases} 1 - x^2, & x \leq 1 \\ 0, & x > 1 \end{cases}$. Hence evaluate $\int_0^\infty \left(\frac{x \cos x - \sin x}{x^3} \right) \cos \frac{x}{2} dx$.	6
C)	Evaluate the integral $\int_0^\infty \frac{t^2}{(t^2+1)^2} dt = \frac{\pi}{4}$.	6
Q.4	Solve any Two of the following:	12
A)	The partial differential equation by eliminating the arbitrary function from $z = x + y + f(xy)$	6
B)	The partial differential equations by eliminating the arbitrary constant $z = (x^2 + a)(y^2 + b)$	6
C)	Solve the following partial differential equations $(mz - ny)p + (nx + lz)q = ly - mx$ where the symbols have got their usual meanings.	6
Q. 5	Solve any Two of the following:	12
A)	Show that $u = x^3 - 3xy^2 + 3x^2 - 3y^2 + 1$ is a harmonic function and hence determine the corre-	6

	sponding analytic function	
B)	If $f(z)$ is an analytic function with constant modulus, show that $f(z)$ is constant	6
C)	Under the transformation $W = \frac{1}{z}$, find the image of $ z - 2i = 2$.	6
Q. 6	Solve any Two of the following:	
A)	Evaluate $\int_0^{1+i} (x^2 + iy) dz$ along the path $y = x$ and $y = x^2$	6
B)	Evaluate $\oint_C \frac{e^{-z}}{z+1} dz$ where C is the circle $ z = 2$ and $ z = \frac{1}{2}$	6
C)	Use Cauchy's integral formula to evaluate $\oint_C \frac{e^{2z}}{(z+1)^4} dz$, where C the circle is $ z = 2$.	6
	*** End ***	